

Lesson Notes

MODULE 7 | LESSON 1

VALUATION IN NORMAL AND STRESSED MARKETS

Reading Time	90 minutes
Prior Knowledge	Comprehensive from Modules 1 through 6
Keywords	Bond Valuation, Stock Valuation, Option Valuation, ETFs, Risk

In the previous modules, we were introduced to several key themes. In this lesson, we will review the valuation of bonds, stocks, options, and ETFs in normal markets. Then, we will understand how these valuations change in light of stresses: changing rates, extreme events, and underestimated volatilities, as well as sector hits. In the case of bonds we devote a good amount of attention to the concepts of duration and convexity which are very important when working with fixed-income securities.

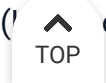
1. Bond Valuation in Normal Markets

Let's begin with bonds. Recall that bonds followed the discounted cash flows. Pricing a bond essentially involves finding the single yield, known as the yield to maturity, by which all cash flows are discounted. For bonds, those coupon sizes are fixed, hence the term fixed-income market, so this becomes a repeated exercise in discounting.

The industry throws an extra complexity in how it calculates time. There are day count conventions that measure how many days have elapsed since the last coupon was paid and how many days in total there are in a period. Not always intuitive, some of these day-count conventions do not use the actual number of days but rather a simplified counting system (e.g., 30 days in each and every month). For now, we will neglect day count conventions and instead focus on simply periods.

Suppose we have a 3% bond that has a 2-year maturity. In total, there are 5 cash flows. The yield-to-maturity is the number that makes the discounted cash flows match the market price. Maturity and coupon are known. Price and yield determine each other.

What if we don't know the size of the cash flow? Well, in the world of fixed income, that is not an issue. Indeed, this is where fixed income gets its name. These periodic payments are fixed in timing and size. (We do not consider floating rates bonds here). The reliability of a fixed coupon makes bonds a relatively



steady and safe investment. But what happens if we relax that assumption? In the next section, we'll cover stocks.

2. Stock Valuation in Normal Markets

We are now going to look at how to value a stock. There are several models to value a stock: discounted cash flow model, relative valuation, dividend discount model, etc. Here, we will consider the dividend discount model, which is predicated upon a company paying dividends.

Earlier you learned about growth stocks—that is not the subject here. Instead, we focus on value stocks, which are stocks that provide income. How do these compare to bonds? Well, bonds are required to pay their coupons or they go into default. Stocks are not required to pay a dividend, even value stocks, but historically have paid dividends consistently. If a value stock failed to pay a dividend, or even pays a lower-than-expected dividend, indeed, the stock price takes a hit. This is often why companies report their earnings and dividends after the market closes; otherwise, the sudden and surprising news could cause undesirable volatility. Clearly, then the dividend yield is crucial in valuing the stock.

What are the two key differences between a bond and a stock?

1. The bond has a specified, agreed-upon maturity date. At that date, the final coupon is paid and the notional amount reimbursed. After that date, no further cash flows occur. Stocks, on the other hand, do not have a "maturity" or an "expiration" date. Companies are theoretically meant to last forever. Therefore, the dividend stream of a stock is never supposed to end.
2. Second, bonds have a specified coupon size. Stocks do not. The size of the dividend is random. If a company does not earn profits in a given period of time and does not have retained earnings, or simply decides not to pay a dividend, there is no dividend. Therefore, dividend sizes are random variables.

To summarize, bonds have 2 known parameters: time to maturity and cash flow size. Stocks have neither of these. The problem of not having a time to maturity is solved by realizing the stock does not have one. The question of cash flow size is answered by modeling the size as a constant or variable. The variable itself can be modeled as one that grows at a constant rate or dynamic rate. We simplify the first by assuming there is no end period. How do we simplify the second? We may assume a constant dividend, a dividend that grows at a constant rate, or a dividend that grows at an increasing rate.

The constant dividend model is flawed if for no other reason than inflation. A company failing to pay a higher dividend is seen as not keeping up with inflation. If these were the model we used, then it would be very similar to fixed income valuation—except that we somehow have to handle the perpetual nature of the cash flows. When we value equities, we'll see that it pays to consider a dividend that grows at a constant rate or even one that grows at a variable rate.

3. Option Valuation in Normal Markets

Stocks and bonds are fundamental assets in the market. Derivatives are derived from these fundamental securities. When we value stocks and bonds, we are using cash flows. When we value

derivatives, we tend to use replication arguments, which use stocks and bonds. Think of stocks and bonds as atoms. Yes, they have simpler parts (like protons and electrons). Think of derivatives as compounds containing atoms in different amounts. Indeed, it will be shown later in the program that an option can be thought of as a combination of a stock and a bond. You'll have to wait for Derivative Pricing to see the details, but for now, let's review the basics of option valuation in normal markets.

Suppose we want to value a call option whose underlying is a stock. We saw in a previous module that there were 6 key inputs. The first 5 of these are very easy to obtain from the market: the stock price, the strike level, the time to expiration, the risk-free rate, and the dividend yield. The final input, the volatility, is not as easy to observe. We need a model for it. One model we may use is the standard deviation of return over the last 30 days, or we may use a time-series model that recognizes volatility is mean-reverting and use that long-term (ergodic) volatility as the estimate for the volatility. Alternatively, we may simply use a proprietary model. When we put volatility in, we get a price.

Suppose we get volatility wrong. In this course, we saw that one problem with volatility is that it may not be constant. We examined distributions that we tested for normality, and then we showed the assumptions required for the normal distribution. Specifically, the kurtosis showed that the volatility itself seemed to have uncertainty with it. As a result, we may input a volatility whose value is less than the true volatility. In this sense, we underestimate volatility in normal markets. Nevertheless, a standard deviation may work well. That value becomes our input in pricing the option. In normal (usual) markets, hopefully, this volatility works well. But if we get this wrong, we underprice options: a costly mistake if we sold them and are hedging them to the tune of high volatility.

4. ETF Valuation in Normal Markets

A portfolio has combinations of securities. One way to estimate the return is to find the weights of each of the securities and the expected returns of each of the securities. Fortunately, return is linear, so we can simply take the weighted average of the returns. Using those returns, we can calculate a covariance matrix. The covariance matrix and historical returns can provide an expected measure of future volatility. If the covariance matrix refers to normal markets, then the covariance matrix should indicate future volatility. Of course, the markets may not remain normal but may become turbulent. To what extent would the covariance matrix degrade?

There are two ways the covariance matrix may change significantly. One is that variances get substantially higher. This would result in higher volatilities along the diagonal of the covariance matrix. Here, we have the same problem as in the previous section—the market may exhibit higher volatilities than we anticipated. Second, the correlations may increase, which means we have less diversification than expected.

We will have to wait until Econometrics to discuss issues like stationarity, which relate to variances becoming more or less constant over time.

5. Market Stress



There are several things that can frazzle interest rates. In many countries, the central bank can set policy to raise or lower interest rates, in an effort to combat high unemployment and/or high inflation. Indeed, interest rate hikes and cuts can send other asset classes into a frenzy. For example, in the first half of 2022, not only interest rate hikes but also the acceleration at which those hikes were expected sent equities reeling, causing a stock market correction into bear territory.

Modeling interest rates by merely looking at a five- or 10-year history may not take into account older time periods that are more relevant to the central bank phases of expansion and tightening. More recent time series do not necessarily equate to more relevant data.

5.2 Equity Markets

In normal markets, the returns of stocks are shown to be fairly normal or Gaussian. In highly volatile markets, stock returns may still be Gaussian, but they may have much higher volatilities. Worse, the distributions may look more like a mixture of normals, with excess kurtosis and even jumps in the stock prices that increase the likelihood of an extreme event occurring.

5.3 Option Markets

In normal markets, options are modeled by careful calculation of the volatility of the underlying, the key input into pricing options in the first place. In turbulent markets, volatilities undoubtedly rise. Will we be able to price and hedge options correctly in a world where volatility reaches higher levels and rises and falls more quickly? Indeed, we can see volatilities easily exceed 100% for many equities and other asset classes in these markets.

5.4 Sector Markets

The issue of diversification that we thought was present may not actually be there. Suppose we are exposed to a sector that crashes. We may earn profits in multiple sectors but lose money in others. However, in highly volatile markets, correlations can rise, which lowers diversification. The promise of diversification to help hedge against a downturn turned out to be a false sense of security, as the heightened correlation treats the portfolio more like a single stock than a basket.

6. Interest Rate Risk with Duration and Convexity

By now, we are accustomed to associate volatility with risk when dealing with equities. However, bonds are different creatures. While volatility exists even for the price of bonds, bond investors are much more concerned about interest rate risk. And the reason is simple: when interest rate changes, so do the prices of bonds.

A key metric to gauge interest rate risk for fixed income investment is **duration**, which is a measure of the sensitivity of the price of a bond or other debt instrument to a change in interest rates. In general, the higher the duration, the more the price of a bond will drop as interest rates rise (thus we bear a greater the interest rate risk). For example, if rates were to rise 1%, a bond or bond fund with a five-year average duration would likely lose approximately 5% of its value. We will make this more rigorous in the following paragraphs.

NOTE. Making everything somewhat rigorous requires a bit of Calculus, but not much more than a knowledge of first and second order derivatives and Taylor's expansions. Something that is expected of students who plan to get a Master's degree in Financial Engineering. However, derivations/calculations will not be tested and only the use of these formulae will.

To understand how the duration and convexity formulae work, we start by recalling the formula for the price of a bond; i.e., the price of the bond at time t is given by:

$$P_t(r) = \sum_{i=1}^N \frac{CF_i}{(1+r)^{t_i-t}},$$

where CF_i is the cash flow (coupon payments and principal repayments) at time t_i for $i = 1, 2, \dots, N$, and r is the relevant yield to maturity or the market discount rate to use at time t . As you can see, $P_t(r)$ is a function of r and calculated at time t . Then, we can calculate the first and second derivatives of $P_t(r)$ with respect to r , which with some tedious but simple calculations can be shown to be:

$$P'_t(r) = \frac{dP_t(r)}{dr} = - \sum_{i=1}^N \frac{CF_i}{(1+r)^{t_i-t+1}} (t_i - t)$$

and

$$P''_t(r) = \frac{d^2 P_t(r)}{dr^2} = \sum_{i=1}^N \frac{(t_i - t)(t_i - t + 1)CF_i}{(1+r)^{t_i-t+2}}.$$

These formulae may look cumbersome, but they only involve basic arithmetic manipulations of the kind easily performed using a spreadsheet. Alternatively, it is possible to code them in Python or any other programming language.

The formula involving the first derivative, $P'_t(r)$, is known as the **\$-duration** of the bond P , at time t , when the yield to maturity is r . Instead, the second derivative, $P''_t(r)$, is known as the **\$-convexity** of the bond P , at time t when the yield to maturity is r .

How do \$-duration and \$-convexity, play a role in helping us to assess interest rate risk?

Consider initially the case of a first order Taylor's expansion of $P_t(r)$:

$$dP_t(r) = P_t(r + dr) - P_t(r) = P'_t(r)dr + o(dr) \approx \frac{dP_t(r)}{dr}dr \equiv \text{\$-duration} \times dr$$

where $o(dr)$ is a notation to indicate that the additional terms not included go to zero when dr goes to zero, faster than dr does and thus are negligible. Thus, the change in value (dollars or another currency) is approximately equal to $P'_t(r)dr$ plus an infinitesimal term $o(r)$, which can be ignored for small movements in the yield to maturity, r . In other words, the \$-duration measures the change in the price of the bond due to the slope of the relationship between $P_t(r)$ and r , that is it informs an

investor about how the price of the bond changes following a small change of the yield to maturity using a first order approximation.

Now, if we divide both sides of the previous equation by $P_t(r)$, we get

$$\frac{dP_t(r)}{P_t(r)} = \frac{P'_t(r)}{P_t(r)} dr + o(dr)$$

and, if we let

$$\text{MD}_{t,r} \equiv -\frac{P'_t(r)}{P_t(r)}$$

which called the **modified duration** of bond P , then

$\text{MD}_{t,r}$ provides the relative change in value in the price of a bond at time t for a small change in r ; i.e.,

$$\frac{dP_t(r)}{P_t(r)} \approx -\text{MD}_{t,r} \times dr.$$

NOTE. In the quizzes, you will just be given either MD or \$-duration or \$-convexity numbers and you will not have to be concerned about how they were derived. But this is a task often assigned to quantitative analysts and they inform us about interest rate risk for fixed income instruments like bonds.

Summarizing, it is useful to think in terms of profits and losses (P&Ls); i.e.,

$$\text{absolute gain/loss} \approx \$\text{-duration} \times dr$$

and

$$\text{relative/percent gain/loss} \approx -\text{MD} \times dr.$$

NOTE. The last two formulae must be known to answer quizzes!

Let's see the formulae at work with a numerical example.

Example. Consider a bond with a maturity of 10 years, a coupon rate of 6%, and a yield to maturity of 5%. For simplicity, assume that the coupon frequency and compounding frequency are both annual. An analyst has calculated that the bond's \$-duration is -810 and the MD is 7.55. Assume that the yield to maturity decreases by 0.1%. What is the change in price for the bond in dollars and as a percentage?

Answer. The absolute loss/gain can be computed using the \$-duration. In this case, it is a gain and more specifically

$$\text{absolute gain} = -810 \times (-0.1\%) = 0.81$$

i.e the price of the bonds goes up by \$0.81.

In percentage terms, the gain is

$$\text{relative gain} = -7.55 \times (-0.1\%) = 0.755\%.$$

Remark. Note that if the frequency of coupon payments or the compounding frequencies are not annual, the formula for dollar duration is different. It can (easily) be shown that in the semi-annual payments case we have:

$$P'_t(r) = - \sum_{i=1}^{2N} \frac{CF_i(t_i - t)}{(1 + r/2)^{2(t_i - t) + 1}}.$$

However, we are not going there and leave it for the students to derive a general formula in the case of k -payments a year. Aspiring financial engineers need to be able to do that on their own.

As the relationship between the price of a bond P and interest, r , is non-linear, it is necessary to consider also the second derivative of the relationship. This is even truer if we consider changes in the yield to maturity that are not really quite small. This leads us to need to include **convexity**. From a mathematical point of view, we can approximate $P_t(r)$ using a second order Taylor's expansion:

$$dP_t(r) = P'_t(r)dr + \frac{1}{2}P''_t(r)(dr)^2 + o((dr)^2)$$

and, recalling what we did a little earlier, we can write this formula as:

$$dP_t(r) \approx (\$ - \text{duration})_{t,r} \times dr + \frac{1}{2}(\$ - \text{convexity})_{t,r} \times (dr)^2$$

where this time, we are including small terms like $(dr)^2$, but we are ignoring terms of order $o((dr)^2)$, which we consider negligible for our purposes.

$$(\$ - \text{convexity})_{t,r} \times (dr)^2$$

measures the change in the price of the bond due to the curvature of the relationship between $P_t(r)$ and r .

The last formula above, could be easier to remember as follows:

$$\text{absolute gain/loss} \approx \$\text{-duration} \times dr + \frac{1}{2}\$ \text{-convexity} \times (dr)^2.$$

That's enough math for us in this course. Let's review what we learned regarding bonds.

- When the relevant interest rate increases, then the price of the bond decreases. If the interest rate decreases, the price of the bond increases.
- The amount by which the price increases or decreases can be determined using \$-duration (first order approximation) or \$-duration and \$-convexity (second order approximation).
- The formulae provide accurate results only if the change in the interest rates are small. If the change in interest rate is not small, say in the order of a few basis points, then the formulae are in general not accurate and, in any case, using convexity and duration together would provide more accurate approximations.

- It should also be clear that the higher the Δ -duration is, the more sensitive to changes in interest rates the price of the bond is and the more important it becomes to hedge interest rate risk.
- The change in interest rate are assumed to happen in a parallel fashion, i.e. the entire yield curve moves up or down by the same amount (parallel shifts). However, sometimes, the yield curve steepens, flattens, or twists. These non-parallel movements are not handled well by traditional methods of duration. Other methods, including principal components, can be used to provide risk measures of how much variation is caused by tilt, twist, and other types of yield curve movements.

Duration as a concept is more than a century old! It predates computers, spreadsheets, and software, all of which today can easily compute the risks associated with movements of the yield curve that are not parallel. While duration still has its use, when we want to understand the effects that stressed markets can have on bond prices, and fixed income assets in general, we can use duration and other measures to examine the sensitivity of prices to changes in yield.

7. Equity Risk with Heavy Tails

In a healthy market, volatility of a stock would remain constant. Then, its corresponding options can be priced with a constant volatility as if the equity's returns are Gaussian. Of course, markets do not always follow the assumptions we would like them to have. Therefore, the volatility can change in stochastic ways. We may want to treat volatility as a parameter—a constant. But markets tend to show that it behaves like a variable—a time-varying quantity. Variables can produce extreme values, and this gives rise to heavy tails. Perhaps you will hear someone remark, “Wow, this was a big loss for stocks, a 3-sigma move—such an unlikely move.” Keep in mind that they are providing a perspective that views the world as Gaussian. Indeed, in the Gaussian world, 3-sigma events are indeed rare: perhaps we get three events every four years! (99.7% of the time, data lies within three standard deviations. That means only .3% of events are outside that interval. With 250 trading days a year, that equates to about three events every four years). Yet, who says that equity returns have to follow Gaussian distributions? At best, they may be Gaussian, but the volatility simply increased; that is, we underestimated volatility. At worst, they are a more complicated structure, such as non-Gaussian, or a mixture of normal random variables, or even a time-changing distribution that requires regime-switching models or other complicated modeling.

As you read this lesson's required reading “Financial Stability Report,” you'll see that asset valuation depends very much on the compensation investors demand for taking risk—the so-called risk premium. At times, the appetite for risk is so great that there are high asset valuations; that is, stocks are expensive when comparing their price to earnings or similar metrics. In stressed times, that appetite can drop significantly and abruptly, indicating a very different price investors are willing to pay for risk. As the paper indicates, “The risk premium for an asset varies over time and, unlike the price of an asset, cannot be directly observed.” This is just one of many reasons why equity valuation is among the most challenging problems in finance, and perhaps one of the areas with the greatest amount of disagreement. Further discussion of asset valuation can be found in the semi-annual report published by the Federal Reserve.

Equity Risk with Underestimated Volatility

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Following up on the previous section, difficulties in modeling volatility lead to difficulty in pricing options. Volatility could be greater than we think for an untold number of reasons. Let's consider several of them.

First, in a Gaussian world, options can be priced with a constant volatility. We just saw that if volatility is a variable rather than a parameter, we should expect volatility to change over time. So what if we based volatility on a relatively low period of volatility, but then volatility is higher? In this new volatile world, returns could still be Gaussian, but we've underestimated the volatility: a statistical error of inferring too small a value for the statistical estimate.

Second, we may have incorrectly assumed that the stock returns were Gaussian in the first place due to heavy tails: kurtosis. Suppose their distribution is closer to a Student's t-distribution. Then, we would tend to underestimate the frequency of large moves by observing the world with our Gaussian glasses.

Third, we may have incorrectly assumed that the stock returns were Gaussian because we assumed symmetry when in fact there are skewed tails indicating asymmetry. Then, we would tend to underestimate the frequency of a negative return because we didn't take this skewness into account.

Fourth, we may have incorrectly assumed that the stock prices could not jump and therefore assumed that extreme moves could only occur if "3-sigma" or "4-sigma" or similar extreme events occur. The Gaussian distribution, however, shows that these types of events are extremely rare.

Fifth, we may have underestimated the costs and complexities of hedging the position, which adds costs to the volatility itself. In other words, real-world costs of hedging make the implied volatility higher than realized volatilities. That makes option pricing risky and hedging more difficult. At worst, we underestimate the size of volatility and the costs of hedging and sell options into the market too cheaply. Then, the options go into the money, and we find ourselves hedging at considerable expense.

Prudent option writing and buying require us to have a solid grasp of both the complexities of volatility and the dynamics of hedging. For example, hedging a put may require the option writer to sell short the underlying stock—a transaction that requires a financing cost. That financing cost can vary over time and get prohibitively expensive. So not only is there a risk of underestimating volatility, but there is also a risk that we underestimate the cost of hedging.

9. Sector Risk with Factor Exposure

One takeaway from COVID-19 is how fragile the supply chain can be. For example, an automobile can have 99% of its parts readily available, but one of them—perhaps even the microchips—are unavailable due to disruptions in the supply chain. These may arise from the inability to source the materials, new regulation that prohibits importing materials from countries that are not considered "free-trade counterparties," the lack of shipping and trucking to transport goods, etc. The consequences are far-reaching (The Board of the International Organization of Securities Commissions). There can be inadequate supply. In addition, the demand can be pent up—during COVID-19, there was greater demand worldwide for automobiles as people reduced their use of public transportation. You can see how a sector like automobiles can be impacted.

On the other side, business travel was greatly reduced during COVID-19, from regular business travelers to conference attendees. Remote work has become increasingly acceptable for many businesses worldwide. Of course, there are still many people who want to take vacations (holidays) and can still choose to travel, but disruptions to worldwide travel significantly impact the airline and hotel industries.

Sectors can be impacted by a variety of factors, including global pandemics. New regulation, geopolitical risk, and inflation affect sectors and can boost or bust a sector. The classical methods of examining which sectors perform well in specific business cycles need updating to provide better analytics. The key is often to find the underlying factors that affect sectors and test their vulnerabilities.

Remember, though, that good ETFs have strength in numbers: diversification by having pooled idiosyncratic risks. ETFs were shown to be very resilient during the COVID-19 pandemic. See <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD682.pdf>

10. Conclusion

Bonds, equities, options, and ETFs are four of the asset classes we've examined in this course. This introductory lesson compared how their pricing and sensitivity change normal markets compared to stressed markets. In the next lesson, we'll tie these in with the sets of related financial concepts.

References

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